Cambridge (CIE) A Level Maths: Probability & Statistics 1

Your notes

Representation of Data

Contents

- Data Presentation
- Stem & Leaf Diagrams
- Box Plots & Cumulative Frequency
- Histograms

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Data Presentation

What graphs and diagrams should I be familiar with?

- You will be expected to be able to use a variety of graphs such as:
  - **Stem-and-leaf diagrams**
    - Can be used with ungrouped data of a single variable
    - Shows all the data and the shape of its distribution
  - **Box plots**
    - Can be used with ungrouped data of a single variable
    - Shows the range, interquartile range and quartiles clearly
    - Very useful for comparing data patterns quickly
  - **Cumulative frequency graphs**
    - Can be used with continuous grouped data of a single variable
    - Shows the running total of the frequencies that fall below the upper bound of each class
  - **Histograms**
    - Can be used with continuous grouped data of a single variable
    - Can be used with varying group sizes
    - Shows the frequencies of the group, represented by the area of each bar
- You might be expected to draw a full diagram or to add to an incomplete diagram

What should I look out for when interpreting graphs?

- Look carefully at the context of the information given in the graph
- Check the scales on both axes carefully, including units
  - Sometimes the numbers will be abbreviated to fit on the scale, for example if a population is given in millions then the number 60 will represent 60 000 000
  - Look carefully at the labels and units to determine how a value should be read
- If there is more than one graph represented on the same set of axes take extra care to ensure you are reading from the correct one
- Beware of misleading graphs, the scales on the axes, units and representation can be manipulated to make a graph look more/less convincing

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Worked Example

A student is collecting information on his friends' interests and believes that his friends who only have dogs spend more time outside than his friends who only have cats. He has surveyed 20 friends with only cats and 20 friends with only dogs and has written down the total amount of time, rounded to the nearest hour, each of them spent outside last week. Describe, with a reason, which diagram would be best for the student to use to display the data.

- The data is raw and ungrouped
- Time, rounded to the nearest hour, is discrete
- The student wants to compare the distributions

The student should use two box plots to display the data, one for his friends with cats and one for his friends with dogs.

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Examiner Tips and Tricks

- Take the time needed when working with diagrams, they are usually 'easy marks' questions but it is common for students to rush them and make silly mistakes.

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Stem and Leaf Diagrams

What is a stem and leaf diagram?

- A **stem and leaf** diagram shows **ALL RAW** data and groups it into **class intervals**
- Stem and leaf diagrams lend themselves to two-digit data but can be used with three-digit data, rarely more

0	1	3	4	4		(4)
1	0	2	6	7		(4)
2	3	3	3	5	8	(5)
3	1	2				(2)

KEY 2 | 3 REPRESENTS 2.3

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- The numbers in **brackets** indicate how many values are in that class interval
  - These are not always included but can be useful when there is a large amount of data to display

How do I draw a stem and leaf diagram?

- Identify the **stems** and the **leaves**
  - **Leaves** would always be **single** digits
    - the number **2** would be represented by **12 | 2**
- If starting from **unordered** data draw **two** diagrams
  - The **first** diagram should get the data into the right **format**
    - i.e. a list of stems with their corresponding leaves
  - The **second** diagram should have stems and leaves in **order**, with a **key**
  - This helps **accuracy** as values are less likely to be missed out

What are stem and leaf diagrams used for?

- The data is arranged into **classes** so at a glance it is possible to see the **modal class interval**
- As the data is in order the **median**, **quartiles**, **maximum** and **minimum** can be identified easily
  - Check you can do this – find the minimum, maximum, median and upper and lower quartiles from the stem and leaf diagram at the start of this revision note
  - Note that these five values are those needed in order to construct a **box-and-whisker diagram** (box plot)

- **Outliers**, once defined, can be easily identified and removed

What about back-to-back stem and leaf diagrams?

- These are used when it is helpful for the data to be split into two comparable categories such as boy/girl, child/adult, UK/non-UK. Etc

	BOYS					GIRLS					
(2)			4	2	0	1	3	4	4		(4)
(4)	9	6	3	1	1	0	2	6			(3)
(4)	6	5	4	0	2	3	3	3	5	8	(5)
(1)				2	3	1					(1)

KEY 6 | 2 | 3 REPRESENTS 2.6 FOR THE BOYS AND 2.3 FOR THE GIRLS

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- Note that the leaves on the left-hand side of the stems (Boys) increase from the centre outwards

Are there any variations on stem and leaf diagrams?

- There are a few minor variations on stem and leaf diagrams that you may see online or in different textbooks

n = 15							NUMBER OF DATA VALUES
2	3	REPRESENTS 2.3					KEY (AT TOP)
LOW	0.1	0.3					LOW VALUED OUTLIERS
0	4	4				(2)	
1	0	2	6	7		(4)	
2	3	3	3	5	8	(5)	
3	1					(1)	
HIGH	3.2						HIGH VALUED OUTLIERS

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- Some or all the different/extra features in the diagram above may appear
- These differences can be applied to back-to-back stem and leaf diagrams
- With large amounts of data, the stems may be split into **two rows**
  - Every stem will be listed **twice**
  - The **first** row for a **stem** will contain **leaves 0 – 4**
  - The **second** row will contain **leaves 5 – 9**

What might I be asked to do with a stem and leaf diagram?

- You may be asked to **draw** or **complete** a stem and leaf diagram
- Find **statistical measures** – **median, quartiles** and interquartile range in particular
  - From which you may be required to draw a **box-and-whisker diagram**

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- Identify and remove **outliers**
- **Compare** data shown by stem and leaf diagrams (either separate or back-to-back); comment on two things and each should be in both terms of the maths and the context of the question
  - a comment about **average** (use median)

e.g. the girls' **median** of 88% was higher than the boys' median of 65% so on average the girls performed better on the test

- a comment about **variation** (**spread**) (use interquartile range)

e.g. the girls' **interquartile range** of 30% was greater than the boys' 15% so the boys had more consistent scores on the test

- **Analyse** what would happen to statistical measures such as the median and quartiles if a **value changed** or a **new value** were to be added to the data

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Worked Example

The following stem and leaf diagrams show the times taken by some children and adults to complete a level on a computer game.

				Children							Adults		
(3)			9	8	8	1							(0)
(4)		6	4	3	1	2	3	5	7	8	8		(5)
(5)	7	6	5	4	2	3	0	4	7				(3)
(3)			3	0	0	4	0	1	2				(3)

2 | 3 represents a time of 23 seconds

(a) Compare the times taken to complete the level between the children and the adults.

(b) It is later discovered two of the adults' times had been omitted from the diagram – times of 23 and 42 seconds.
Briefly explain whether adding these times would change the adults' median time.

a) To compare, use an average – so medians – and a measure of spread – so interquartile range

	n	Q_1	Q_2	Q_3
Children	15	4th	8th	12th
Adults	11	3rd	6th	9th

	Children					Adults					
(3)		9	8	8	1						(0)
(4)	6	4	3	1	2	3	5	7	8	8	(5)
(5)	7 6	5	4	2	3	0	4	7			(3)
(3)		3	0	0	4	0	1	2			(3)

	Q_1	Q_2	Q_3	IQR
Children	21	32	37	37 – 21 = 16
Adults	27	30	40	40 – 27 = 13

The children had a slightly higher median time of 32 seconds compared to the adults' 30 seconds
This means, on average, the adults completed the level on the computer game quicker than the children
The children also had a greater interquartile range of 16 seconds compared to the adults' 13 seconds
This means the adults' times to complete the level on the computer game were more consistent (less varied)

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b) 23 seconds is less than the adults median of 30 seconds
42 seconds is greater
Therefore there will be no change to the adults' median time

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Examiner Tips and Tricks

- **Accuracy** is important
  - (Lightly) tick off values as you add them to a stem and leaf diagram
  - Check you have the right number of data values in total on your diagram
    - Other checks can include ensuring the median has the same number of values either side of it

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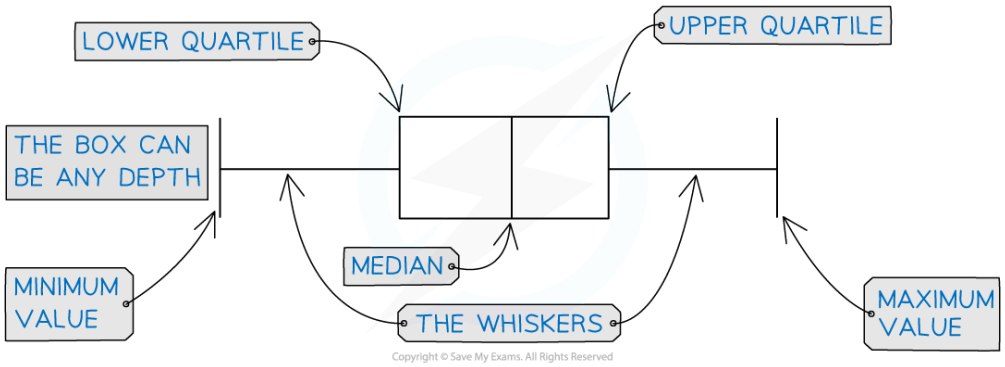
Box Plots

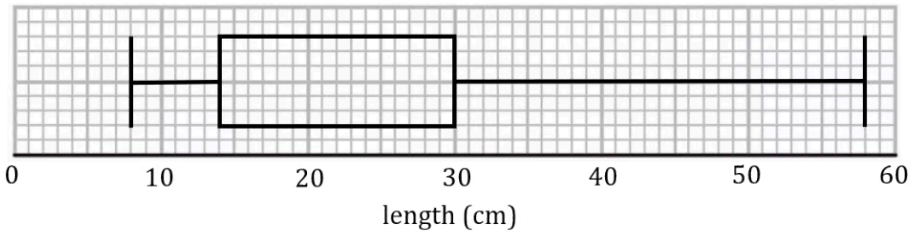
What is a box plot?

- A box plot is a graph that clearly shows key statistics from a data set
  - It shows the **median, quartiles, minimum** and **maximum values** and **outliers**
  - It does not show any other individual data items
- The middle 50% of the data will be represented by the box section of the graph and the lower and upper 25% of the data will be represented by each of the whiskers
- Only one axis is used when graphing a box plot
  - It is still important to make sure the axis has a clear, even scale and is labelled with units
- Box plots are often used for comparing two sets of data
  - Both box plots will be drawn one above the other on the same scale on the x-axis
  - They are useful for **comparing data** because it is easy to see the main shape of the distribution of the data from a box plot



Worked Example

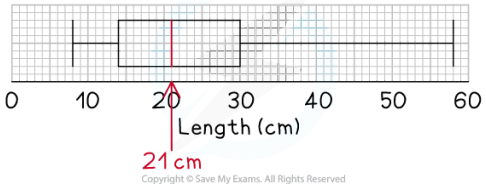
The incomplete box plot below shows the tail lengths in cm of some students' pets.



(i) Given that the median tail length was 21 cm, complete the box plot.

(ii) Find the range and interquartile range of the tail lengths.

(a) Check the scale carefully before plotting the median



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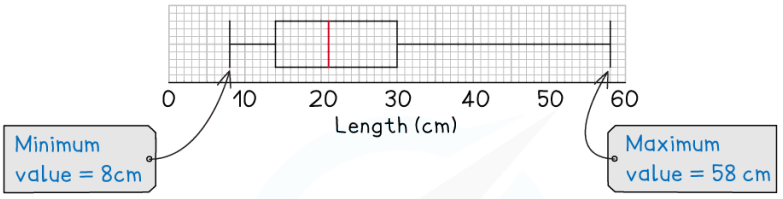
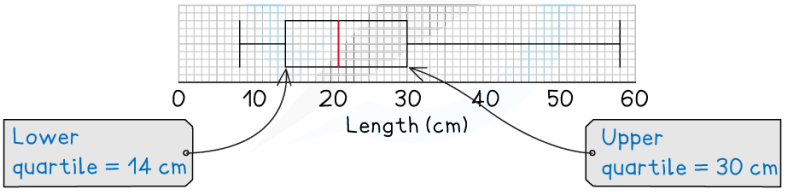
(b) Range = maximum value − minimum value



Range = 58 − 8 = 50 cm

Interquartile range = Upper quartile − Lower quartile



Interquartile range = 30 − 14 = 16 cm

Range = 50 cm, IQR = 16 cm

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Examiner Tips and Tricks

- Remember a box plot is a graph and should be treated like one, even though there is only one axis. It should have a title, a clear, even scale that is labelled with

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units if there are any. If drawing two box plots on the same axis label each one clearly.

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Cumulative Frequency

What is a cumulative frequency graph?

- A cumulative frequency graph is used with data that has been organised into a **grouped frequency** table
- The cumulative frequency graph can be used to find estimates of percentiles and quartiles
  - As the data is grouped, it is not possible to find the actual values of these statistics

What are the main features of a cumulative frequency graph?

- A cumulative frequency graph considers how much data there is up to a certain value, including the data in that group and the one below
- Cumulative frequency will always be plotted on the y – axis
  - Consider the scale carefully because this will usually be a large number
  - You may be asked to add to one or both axes, remember to label both axes clearly and include units on the x – axis if they are needed
- The cumulative frequency is calculated by adding the frequency in each group, or class, to the frequency in the ones before
  - This is essentially accumulating the frequencies as you go
- The cumulative frequency for each class must be plotted against the **upper boundary** of each corresponding class
  - The cumulative frequency that corresponds with each upper boundary will not only consider the frequency of the data in that class, but all of the data in the groups below it too
- When the points have been plotted they should be joined up with a smooth curve
  - However, some may be joined with straight lines from point to point

How do we read statistics from a cumulative frequency graph?

- Quartiles and percentiles can be read from a cumulative frequency graph
  - The median, Q_2 is read from the y – axis scale at the $\frac{n}{2}$ th value
  - The lower quartile, Q_1 ,is read from the $\frac{n}{4}$ th value and the upper quartile, Q_3 is read from the $\frac{3n}{4}$ th value

- Any percentile can be read from the graph by finding the percent of the total frequency and reading from the value on the y - axis
- To read the corresponding data value once the position on the y – axis is known, use a ruler to draw a line from the y – axis to the graph and then down to the x – axis
- Sometimes the frequency of values greater than or less than a particular data value will need to be found, this time you will have to read from the x – axis to the y – axis
  - Take particular care if the question asks for a frequency greater than a particular data value, the value found from the y – axis will need to be subtracted from the total frequency

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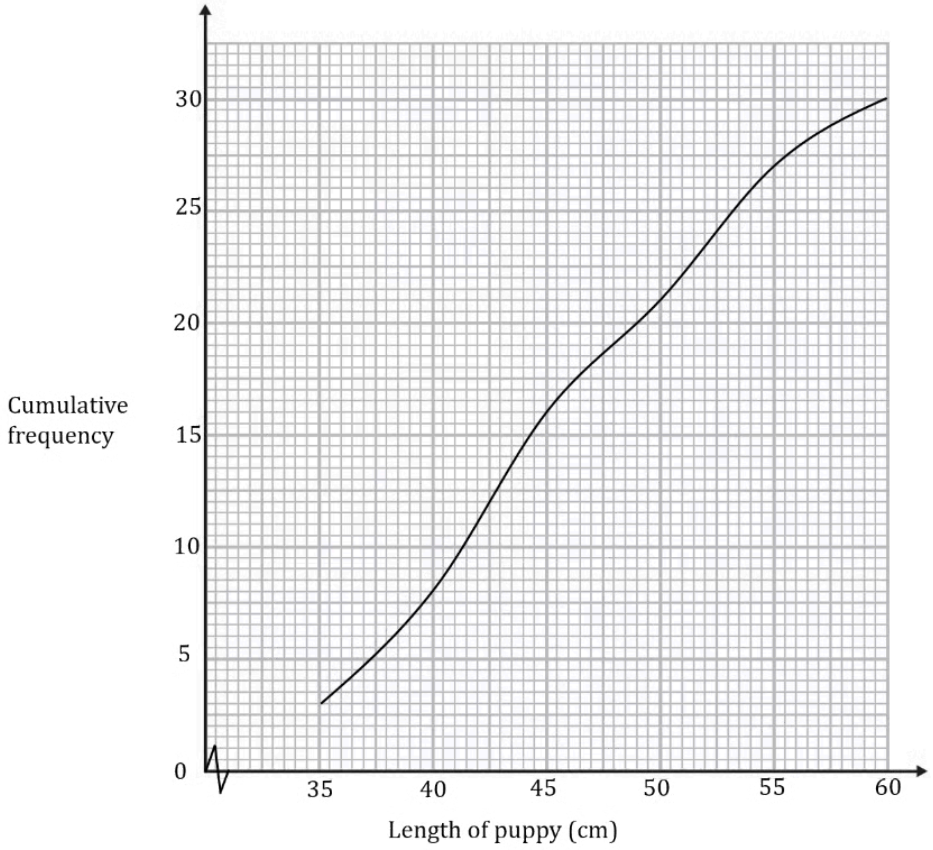
Worked Example

The cumulative frequency graph below shows the lengths in cm, l, of a group of puppies in a training group.

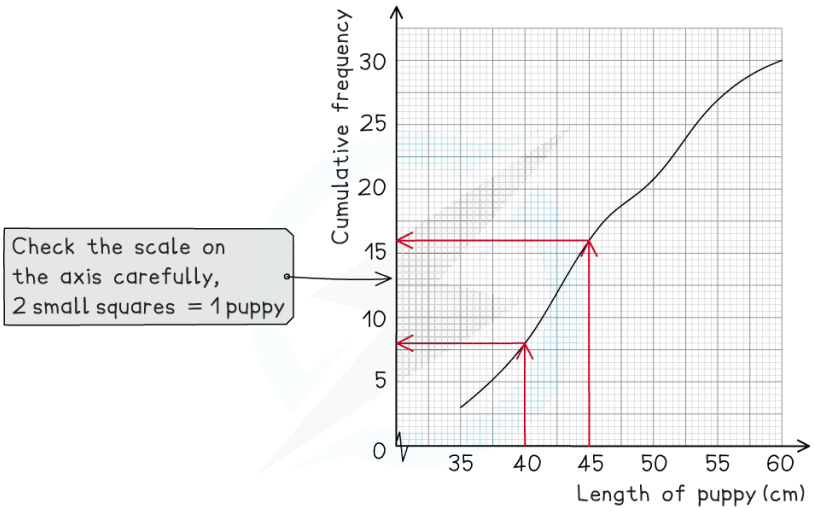


(i) Given that the group $40 < l \leq 45$ was one of the groups used in the data collection, find the number of puppies that were in this group.

(ii) Use the graph to find an estimate for the interquartile range of the puppies.

(iii) X % of the puppies are greater than 53.5 cm long, use your graph to find an estimate for the value of X.

(i) Find the frequency at 40 cm and 45 cm.



8 puppies are $\leqslant$ 40 cm

16 puppies are $\leqslant$ 45 cm

$\therefore 40 < \ell \leqslant 45$ has $16 - 8$ puppies

There are 8 puppies in the group $40 < \ell \leqslant 45$

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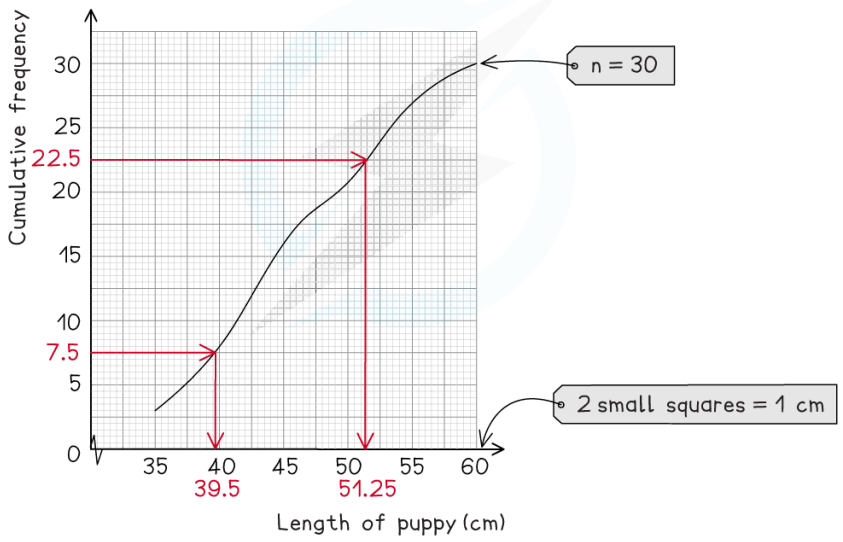
(ii) Interquartile range = upper quartile (Q_3) − Lower quartile (Q_1)

$n = 30 \Rightarrow Q_1 = \frac{30^{th}}{4}$ value $= 7.5^{th}$ value

$Q_3 = 3\left(\frac{30}{4}\right)^{th}$ value $= 22.5^{th}$ value

Can be seen from the top of the graph



IQR = 51.25 − 39.5 = 11.75 cm

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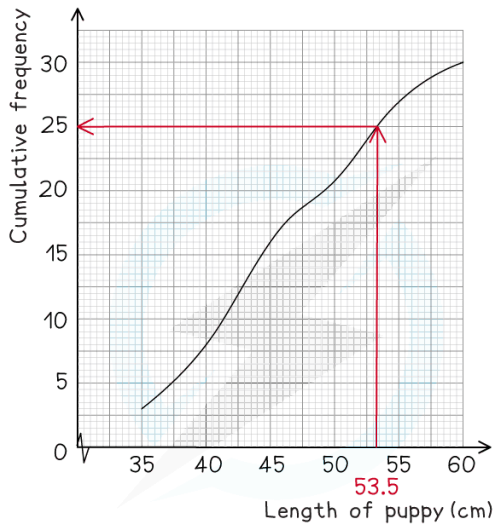
(iii) Read from the x−axis at 53.5 cm.



Approximately 25 puppies are less than 53.5, so 5 must be greater than 53.5 cm.

$\frac{5}{30} = 16.67...\ \%$

$x \approx 16.7\ \%$ (3.s.f.)

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Examiner Tips and Tricks

- If you are asked to read values from your graph make sure you use a ruler and mark the lines on clearly to show where you took your readings from. Remember that the graph shows the accumulated frequencies so if you need only the frequency you may need to subtract the previous value.

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Histograms

What is a histogram?

- A histogram is similar to a bar chart but with some **key differences**
  - A histogram is for displaying **grouped continuous data** whereas a bar chart is for **discrete** or **qualitative data**
  - There will never be any gaps between the bars of adjacent groups in a histogram
  - Whilst in a bar chart the frequency is read from the **height** of the bar, in a histogram the height of the bar is the **frequency density**
- On a histogram **frequency density** is plotted on the y – axis
  - This allows a histogram to be plotted for **unequal class intervals**
  - It is particularly useful if data is spread out at either or both ends
- The **area** of each bar on a histogram will be **proportional** to the frequency in that class

How do I draw a histogram?

- **Step 1.** Always check that there are no gaps between the upper boundary of a class and the lower boundary of the next class
  - If there are gaps you will need to close them by changing the boundaries before carrying out any calculations
    - Consider whether the values are rounded or **truncated** before closing the gaps
- **Step 2.** Find the **class width** of each group by subtracting the lower boundary from the upper boundary
- **Step 3**. Calculate the **frequency density** for each group using the formula:

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

- **Step 4.** The histogram will be drawn with the data values on the x – axis and frequency density on the y – axis
  - Remember that the scale on both axes must be even, although the class widths may be uneven
    - Both axes should be clearly labelled and units included on the x – axis
  - Most often, the bars will have different widths

How do we interpret a histogram?

- It is important to remember that the y – axis does not tell us the frequency of each bar in the histogram

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- The frequency of a class is found by
  - $\text{Frequency} = \text{Frequency Density} \times \text{Class Width}$
- You may be asked to find the frequency of part of a bar within a histogram
  - Find the area of that section of the bar using any information you have already found out

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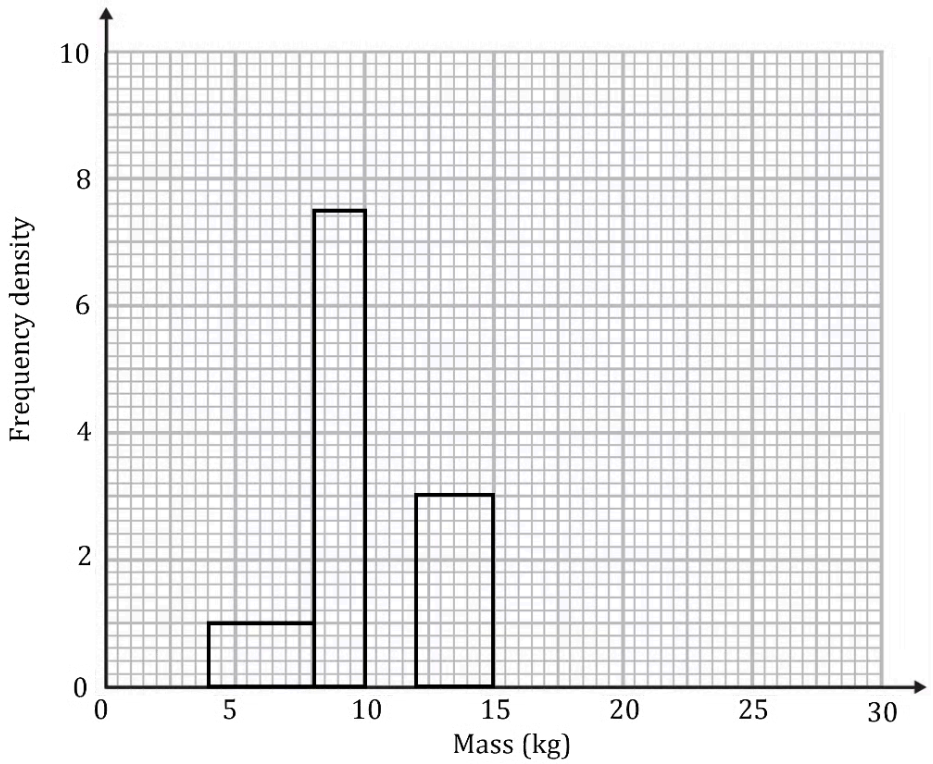
Worked Example

The table below and its corresponding histogram show the mass, in kg, of some new born bottlenose dolphins.

Mass, m kg	Frequency
$4 \le m < 8$	4
$8 \le m < 10$	15
$10 \le m < 12$	19
$12 \le m < 15$	9
$15 \le m < 30$	6



(a) Complete the histogram.

(b) Estimate the number of dolphins whose weight is greater than 13 kg.

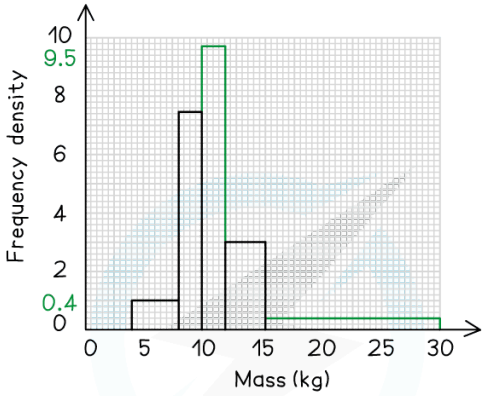
(i) Find the frequency density for $10 \leqslant m < 12$ and $15 \leqslant m < 30$

$$\text{Frequency density} = \frac{\text{Frequency}}{\text{Class Width}}$$

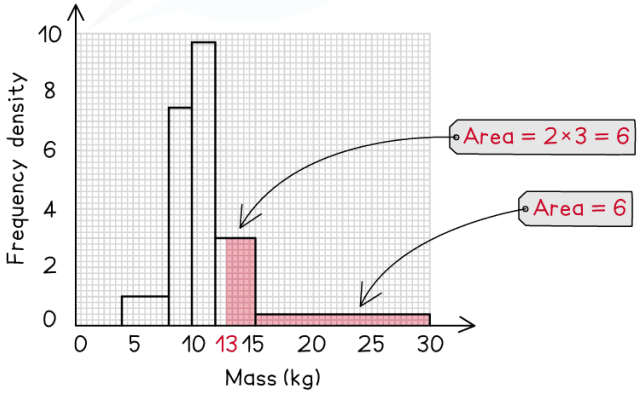
For $10 \leqslant m < 12$ $\quad FD = \frac{19}{2} = 9.5$

For $15 \leqslant m < 30$ $\quad FD = \frac{6}{15} = 0.4$

Draw the bars on the diagram



(ii) We know there are 6 dolphins in the class $15 \leqslant m < 30$
Estimate the frequency of the interval $13 < m < 15$ by finding the area



Total = 6 + 6

An estimated 12 dolphins are greater than 13 kg

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Examiner Tips and Tricks

- Look carefully at the scales on the axes, it will rarely be a simple 1 unit to 1 square.